

■ Research Interests

I am a Ph.D. candidate in computational cognitive science expecting to graduate in early 2027. My doctoral research lies at the intersection of early language acquisition, speech perception, machine learning, and auditory neuroscience. More specifically, I conduct computational simulations of early phonetic learning in fetuses and newborns. Although I have mainly used computational methods in my recent work, I would like to return to empirical research in my next position.

■ Research Experience

Ph.D. Candidate

October 2023 – Present

Aix-Marseille Université – LIS – CRPN – ILCB — Marseille, France

Supervisors : Drs. Thomas Schatz, Claire Kabdebon & Ricard Marxer

- Conducted an extensive review on the prenatal acoustic environment in order to simulate it.
- Modeled fetal auditory learning in a simulated prenatal acoustic environment using machine learning.
- Prepared speech corpora for simulations.

Intern in Auditory Neuroscience

February – June 2023

Laboratoire d'Informatique et Systèmes — Marseille, France

Supervisors : Drs. Thomas Schatz, Clément François & Benjamin Morillon

- Pre-processed EEG datasets.
- Conducted Temporal Response Function (TRFs) analysis on EEG datasets.

Intern in Experimental Psycholinguistics

January – June 2022 (part-time)

Laboratoire Parole et Langage — Aix-en-Provence, France

Supervisor : Dr. Cheryl Frenck-Mestre

- Internship report: *Perception of second language phonemic contrasts in lexical context: an event-related potentials study.*
- Recruited and tested participants for an EEG experiment.

Intern in Natural Language Processing

May – June 2021

Laboratoire Parole et Langage — Aix-en-Provence, France

Supervisor : Professor Nùria Gala

- Conducted a pilot study to assess the linguistic complexity of a corpus of health-related documents.

■ Education

Ph.D. in Computer Science

2023 – Present

Aix-Marseille Université – LIS – CRPN – ILCB — Marseille, France

Supervisors: Drs. Thomas Schatz, Claire Kabdebon & Ricard Marxer

M.S. in Cognitive Science - Language, Communication and the Brain

2021 – 2023

Aix-Marseille Université — Marseille, France

Thesis: *Using naturalistic speech EEG recordings to test the cognitive reality of representations suggested by a Computational model of early phonetic learning.* [Manuscript](#) 

Thesis supervisor: Dr. Thomas Schatz

B.A. in Linguistics

2018 – 2021

Aix-Marseille Université — Aix-en-Provence, France

Minor: Teaching French as a Foreign Language.

■ Peer-Reviewed Publications

Balleroy, A., Kabdebon, C., Marxer, R., & Schatz, T. (In press). The Fetal Acoustic Environment may be too Noisy for Prenatal Phonetic Learning from External Sources. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 48, Jul 2026, Rio de Janeiro, Brazil.

■ Preprints and Manuscripts in Preparation

Kepinska, O., **Balleroy, A.**, & Schatz, T. (2026). A review on the effect of the maternal voice on early language acquisition [Manuscript in Preparation – Working Title].

■ Oral Presentations

Balleroy, A. (2026). *The Fetal Acoustic Environment may be too Noisy for Prenatal Phonetic Learning from External Sources* [Remote Conference Talk]. *Annual Meeting of the Cognitive Science Society*, Rio de Janeiro, Brazil.

Balleroy, A. (2026). *The Fetal Acoustic Environment may be too Noisy for Prenatal Phonetic Learning from External Sources* [Local Invited Talk]. *Institute of Language, Communication and the Brain Docs and Postdocs Workshop*, Marseille, France.

Balleroy, A. (2025). *Modeling the effect of the prenatal acoustic environment on early phonetic learning* [Invited Talk]. *Computational Neuroscience Seminars*, University of Maryland, College Park, United States.

Balleroy, A. (2025). *Modeling the effect of the prenatal acoustic environment on early phonetic learning* [Poster Presentation]. *Society for Research in Child Development Biennial Meeting (SRCD)*, Minneapolis, United States. [Poster](#) .

Balleroy, A. (2025). *Modeling the effect of the prenatal acoustic environment on early phonetic learning*. [Local Invited Talk]. *Institute of Language, Communication and the Brain Docs and Postdocs Workshop*, Marseille, France.

Balleroy, A. (2024). *What is the nature of early speech sounds representations? A large-scale computational modeling study*. [Local Invited Talk]. *Institute of Language, Communication and the Brain Docs and Postdocs Workshop*, Marseille, France.

■ Technical Skills

- **Programming** Python , R 
- **Machine Learning** Scikit-Learn
- **EEG Processing** MNE Python
- **Sound and Signal Processing** Librosa, Praat, Scipy, Torchaudio
- **Data Visualization** Matplotlib, Seaborn, Plotnine, GGplot
- **Data Science** Pandas, Numpy, Jupyter, Doit
- **Project Management and Planning** Reliable, Taiga.io, SuperProductivity
- **High-performance Computing** Slurm
- **Version Control** Git , Github 
- **Scripting** Bash 
- **Statistics** R , Statsmodels, Pingouin-Python
- **Experimental Design** Psychopy
- **Research Methods** Machine Learning, EEG, Reaction Time Data
- **Writing & Literature Review** LaTeX , Overleaf, Zotero, Obsidian
- **Brain-computer Interfaces** Arduino, OpenBCI
- **Scientific Data Extraction** WebplotDigitizer

■ Teaching and Supervision Experience

Teaching Assistant

October 2024 – August 2025

Computer Science Department, Aix-Marseille Université — Marseille, France

- Bachelor level classes: Introductory Data Science (52h), Introductory Machine Learning (12h).
- Taught lab classes, corrected and created assignments, supervised students' projects, managed schedule.

Associate Supervisor

2025

Aix-Marseille Université — Marseille, France

Student: Encina Robles Paula — Erasmus Intern, M.S. in Psychology

■ Service

- Co-review with supervisor for the *Proceedings of the Annual Meeting of the Cognitive Science Society*.

■ Other Relevant Experience

Cognitive Science Consultant for an Upcoming Podcast 2026 – Present
Association Autigirls — Marseille, France

Science Communicator in High Schools and Middle Schools 2024
Genre de Science Project – Labo Sociétal – Ecole Centrale Méditerranée — Marseille, France

■ Fellowships, Grants and Awards

Travel Grant to attend the SRCD Biennial Meeting + Research stay at the University of Maryland – 3050€ 2025
ILCB, LIS, Allocation Mobilité Internationale ED 184, Qarma Team

3-years ILCB PhD Fellowship 2023
Institute of Language, Communication and the Brain

■ Professional Affiliations

Cognitive Science Society, member June 2026 – Present
 Society for Research in Child Development, member May 2025 – May 2026

■ Professional Development / Trainings

Teaching Assistant Training (25h), Center for Innovations in Pedagogy, Aix-Marseille Université 2025

Mental Health First Aid Certificate (14h), PSSM France May 2025

Computational Neuroscience Online Summer School (3 weeks), Neuromatch Academy July 2024

Scientific Integrity in Research MOOC (10h), FUN Campus June 2024

Open Science MOOC (3h) June 2024

Ethics for Research MOOC (10h), FUN Campus June 2024

Introduction to R Software, Statistics, and Multivariate Analysis MOOC (10h) June 2024

Gender Equality in Academic and Professional Settings: The Effects of Gender Stereotypes (6h) March 2024

Strategic Communication: Making Your Research Accessible to Your Target Audience (16h) January 2024

Preventing Violence and Harassment (6h) December 2023

Experimental Methodology for Computer Science (24h) November 2023

DESU Data Science Applied to Neuroscience (3 weeks), Institut Neuromarseille July 2023

Institute of Language, Communication and the Brain Summer School (1 week) 2021, 2023

TOEIC@ Listening and reading test, 975 points, ETS EMEA January 2021

■ Language Skills

- French Native
- English C2
TOEIC@ Listening and reading test, 975 points
- Italian B2
- Japanese B1
- German A2